

TrES-5b

R-1611

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-5_130928 · created 2026-08-02T03:22:48+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2456563.8314 +/- 0.0015 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.184 +/- 0.018
Transit depth (Rp/Rs)^2	3.4 +/- 0.65 [%]
Orbital Inclination (inc)	86.35 +/- 1.97
Airmass coefficient 1 (a1)	1.263 +/- 0.035
Airmass coefficient 2 (a2)	-0.138 +/- 0.013
Scatter in the residuals of the lightcurve fit is	2.77 %
Best Comparison Star	None
Optimal Aperture	2.78
Optimal Annulus	4.68
Transit Duration (day)	0.084 +/- 0.0071

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.184 $\pm$ 0.018 vs 0.142 (deviation 2.19 σ)
Duration fit vs published	0.084 d vs 0.0758 d (deviation 1.08 σ)
Mid-time O-C	-0.5 min
Depth SNR	9.6
Residual scatter	2.77 %

Light curve

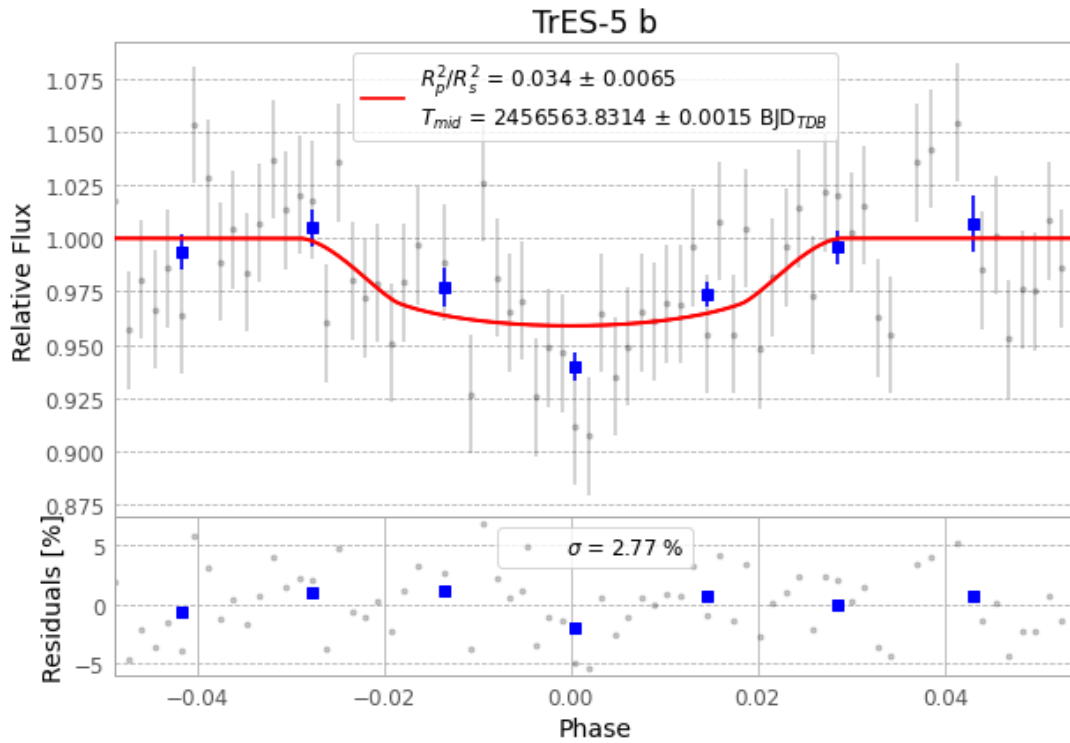


Figure 1 — FinalLightCurve_TrES-5 b_2013-09-27.png (SHA-256 5745d1a4a844813b...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [341, 283], 1 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 271190d461e56cd0...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

74 FITS frames (49 MB), TRES-5130928060912.FITS → TRES-5130928094812.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-5-130928.log (eca343b49a63...), FinalLightCurve_TrES-5 b_2013-09-27.png (5745d1a4a844...), FinalLightCurve_TrES-5 b_2013-09-27.csv (31b43309774a...)

Compute resources

Wall time 100.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 03:22Z.